Statistical Foundations: The Mean and Its Properties

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1 Probability

For an experiment with a finite number of equally likely outcomes, the probability of an event is the ratio between the number of outcomes favorable to the event and the total number of possible outcomes.
More generally, probability theory provides a mathematical framework for assigning probabilities to events. We distinguish two basic concepts:

- **an experiment** is a procedure that produces an outcome;
- **an event** is a set of possible outcomes of the experiment.

A mathematical formulation is the following. Let Ω be the sample space, i.e. the set of all possible outcomes, and let $\mathcal{A}$ be a collection of events (a σ-algebra of subsets of Ω). A probability measure $\mathcal{P}$ assigns a value between 0 and 1 to each event $A \in \mathcal{A}$, with

$$\mathcal{P}(\Omega) = 1.$$

For disjoint events, probabilities are additive:

$$\mathcal{P}\left(\bigcup_i A_i\right) = \sum_i \mathcal{P}(A_i).$$

A simple example is the roll of a fair six-sided die:

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Here, each elementary outcome has probability $1/6$. Events are subsets of Ω, for example

$$A = \{2, 4, 6\},$$

the event that the die shows an even number, for which

$$\mathcal{P}(A) = \frac{3}{6} = \frac{1}{2}.$$

2 Random variables

A random variable is a variable whose value, taken from a given distribution, is determined by the outcome of a random experiment. The randomness means that:

- we do not know the outcome before performing the experiment;
- if we repeat the experiment under the same conditions, the outcome need not be the same.

Mathematically, a random variable X is a *function* from the sample space Ω to $\mathbb{R}$:

$$X : \Omega \to \mathbb{R}.$$

In other words, a random variable assigns a numerical value to each possible outcome in Ω.
We distinguish between **discrete** and **continuous** random variables. A discrete random variable takes values in a finite or countably infinite set. For example, the number of inhabitants of a city is a discrete variable.
A continuous random variable can take any value in an interval of the real line. For example, the height of a person can be considered a continuous variable.

3 Distribution

The distribution of a random variable describes how probability is allocated among its possible values. It therefore tells us the probability that the random variable takes a given value or falls within a given interval.

Discrete random variables
A discrete random variable takes values in a finite or countably infinite set of distinct values. Its distribution is described by a probability mass function, which assigns a probability to each possible value.
The general form is

$$P(X = x_i) = p_i, \qquad \sum_i p_i = 1.$$

A classical example is the roll of a fair die. The random variable X, representing the result of the roll, can take the values $1, 2, 3, 4, 5, 6$, each with probability $1/6$.
Examples of discrete random variables include Bernoulli and binomial random variables.

Continuous random variables
A continuous random variable can take any value in an interval of the real line. Its distribution is described by a probability density function $f(x)$.
For a continuous random variable, the probability of any single exact value is zero:

$$P(X = x) = 0.$$

Probabilities are instead obtained over intervals:

$$P(a \le X \le b) = \int_a^b f(x)\,dx, \qquad \int_{-\infty}^{+\infty} f(x)\,dx = 1.$$

A classical example is human height, which can be approximated by a continuous random variable. Normal and uniform distributions are examples of continuous distributions.

4 Mean and expected value

4.1 Mean

Let X denote the attribute being studied.
For a finite population of N individuals, the population mean, denoted by μ, is the arithmetic mean of the N population values:

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}.$$

The distinction between a finite population and a random variable is important. The population mean above is a descriptive quantity calculated from the values in a finite population. For a random variable, the corresponding theoretical quantity is its expected value.

4.2 Expected value

The expected value, denoted by $E(X)$, is defined differently for discrete and continuous random variables.
For discrete random variables

$$E(X) = \sum_i x_i \, P(X = x_i),$$

where $P(X = x_i)$ is the probability that X takes the value x_i.
The expected value is therefore a weighted mean, with the probability of each possible value providing the weight. For a discrete random variable whose possible values are equally likely, the expected value is the arithmetic mean of those values.
For example, for a fair six-sided die,

$$P(X = x_i) = \frac{1}{6}, \qquad i = 1, \ldots, 6,$$
$$E(X) = \sum_{i=1}^{6} x_i P(X = x_i) = \frac{1 + 2 + \cdots + 6}{6} = 3.5.$$

For continuous random variables, the probability $P(X = x)$ is zero for every exact value x. The probability density $f(x)$ must therefore be used:

$$E(X) = \int_{-\infty}^{+\infty} x f(x) \, dx.$$

For a uniform distribution on a bounded interval, $f(x)$ is constant within that interval and zero elsewhere. For a normal distribution,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right].$$

Here σ is the standard deviation.
A useful interpretation of the expected value is as a long-run average: under repeated independent repetitions of the same random experiment, the sample average tends to the expected value under suitable conditions.

4.3 Linearity properties of the expected value

Let X be a random variable with mean μ_X, Y a random variable with mean μ_Y, and a a constant. Then

$$E(X) = \mu_X,$$
$$E(a + X) = a + \mu_X,$$
$$E(aX) = a\mu_X,$$
$$E(X + Y) = \mu_X + \mu_Y.$$

The last property does **not** require X and Y to be independent.

5 Variance

5.1 Definition

The mean describes the centre of a distribution, but it is not sufficient to describe its spread. A second quantity is therefore needed to quantify the dispersion of values around the mean.
The population variance is denoted by σ^2.
For a random variable X with mean μ, the variance is defined as the expected squared deviation from the mean:

$$Var(X) = E\left[(X-\mu)^2\right].$$

For discrete random variables

$$Var(X) = \sum_i (x_i - \mu)^2 \, P(X = x_i).$$

For continuous random variables

$$Var(X) = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x)\, dx.$$

An equivalent and often useful expression is

$$Var(X) = E(X^2) - [E(X)]^2.$$

For a population, $Var(X) = \sigma^2$.

5.2 Properties of the variance

Let X be a random variable with mean μ_X and variance σ_X^2, Y a random variable with mean μ_Y and variance σ_Y^2, and a a constant. Then

$$\begin{aligned} Var(X) &= \sigma_X^2, \\ Var(a+X) &= \sigma_X^2, \\ Var(aX) &= a^2\sigma_X^2, \\ Var(X+Y) &= \sigma_X^2 + \sigma_Y^2 + 2Cov(X,Y), \end{aligned}$$

where

$$Cov(X,Y) = E\left[(X - E(X))(Y - E(Y))\right].$$

If X and Y are independent, then $Cov(X,Y) = 0$, and therefore

$$Var(X+Y) = \sigma_X^2 + \sigma_Y^2.$$

6 Notable values for classical distributions

For a Bernoulli random variable, $X = 1$ for a success and $X = 0$ otherwise, and p is the probability of success. Then

$$E(X) = 1 \cdot p + 0 \cdot (1-p) = p,$$

and

$$Var(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1-p).$$

A binomial random variable represents the number of successes in n independent Bernoulli trials with the same success probability p. It can therefore be written as a sum of n independent Bernoulli random variables:

$$X = \sum_{i=1}^{n} X_i.$$

Consequently,

$$E(X) = np, \qquad Var(X) = np(1-p).$$

7 Convergence

7.1 Central Limit Theorem

In a simplified form, the Central Limit Theorem states that the standardized sum (or, equivalently, the standardized sample mean) of a large number of independent and identically distributed random variables with finite variance approaches a standard normal distribution.
If $X_1, \ldots, X_n$ are i.i.d., with mean μ and finite variance σ^2, then

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

Thus, for sufficiently large n, the sampling distribution of the mean can often be approximated by a normal distribution, even when the original variables are not normally distributed.
The theorem is about the **distribution of the standardized sum or sample mean**; it does not state that the individual observations become normally distributed.

7.2 Law of Large Numbers

Let $X_1, \ldots, X_n$ be independent observations from a population with mean μ, and let

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}.$$

The Law of Large Numbers answers the question: what happens to $\bar{X}$ as n becomes large?
Under the usual conditions,

$$\bar{X} \xrightarrow{\text{a.s.}} \mu.$$

In words, the sample mean converges to the population mean as the sample size increases.
For example, if $X_i \sim Bern(p)$ independently, then

$$\frac{X_1 + X_2 + \cdots + X_n}{n} \xrightarrow{\text{a.s.}} p.$$

The Law of Large Numbers is a statement about convergence. Separately, by linearity of expectation,

$$E(\bar{X}) = \mu,$$

so the sample mean is an unbiased estimator of the population mean.
The question of what constitutes a "sufficiently large" sample depends on the distribution and on the purpose of the analysis. There is no universal rule that $n = 30$ is always sufficient.

8 Normal distribution

The normal distribution is a continuous probability distribution with many important applications in statistics. It also plays a central role in statistical inference because of the Central Limit Theorem: sums and means of many independent observations are often approximately normally distributed after standardization.
The normal distribution is symmetric around its mean. If

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

then

$$X - \mu \sim \mathcal{N}(0, \sigma^2),$$
$$\frac{X - \mu}{\sigma} \sim \mathcal{N}(0,1).$$

The transformation

$$Z = \frac{X - \mu}{\sigma}$$

is called standardization: the variable is centred by subtracting its mean and reduced by dividing by its standard deviation. Standardized normal variables are usually denoted by Z.

This transformation allows us to calculate probabilities. Let

$$X \sim \mathcal{N}(24, 4),$$

where the second parameter is the variance, so that $\sigma = 2$. For a value x drawn from this distribution,

$$P(x > 28) = P\left(\frac{x - 24}{2} > \frac{28 - 24}{2}\right) = P(Z > 2).$$

Since approximately 2.5% of a standard normal distribution lies above $z = 2$,

$$P(Z > 2) \simeq 0.025.$$

For more general critical values, $P(Z > z_{\text{critical}})$ can be evaluated in R with the function `pnorm()`:
(The following three expressions are equivalent:)

```
1 - pnorm(2)
pnorm(2, lower.tail=FALSE)
pnorm(28, mean=24, sd=2, lower.tail=FALSE)
```

9 Confidence interval

The construction of a confidence interval for a population mean relies on the sampling distribution of the sample mean.
For an i.i.d. sample with finite variance,

$$E(\bar{X}) = \mu, \qquad Var(\bar{X}) = \frac{\sigma^2}{n}.$$

The Central Limit Theorem tells us that, for sufficiently large n, the standardized sample mean is approximately normally distributed:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1).$$

When the population is normally distributed, this normal distribution for the standardized mean is exact for every n.
Suppose first that σ is known. We seek a quantity ε_α such that

$$\begin{aligned} P\left(|\bar{X} - \mu| \leq \varepsilon_\alpha\right) &= 1 - \alpha \\ &= P\left[-\varepsilon_\alpha \leq \bar{X} - \mu \leq \varepsilon_\alpha\right]. \end{aligned}$$

Using the standardized normal variable,

$$P\left(|\bar{X} - \mu| \leq \varepsilon_\alpha\right) = P\left[\frac{-\varepsilon_\alpha\sqrt{n}}{\sigma} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{\varepsilon_\alpha\sqrt{n}}{\sigma}\right] = 1 - \alpha.$$

Therefore,

$$\varepsilon_\alpha = z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

The corresponding two-sided confidence interval for μ is

$$\bar{X} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

For example, for a 95% confidence interval, $\alpha = 0.05$ and $z_{0.975} \simeq 1.96$.

10 Student's t distribution

The standardization of the sample mean uses the variance of the sample mean:

$$Var(\bar{X}) = \frac{\sigma^2}{n}.$$

The problem is that σ^2 is generally unknown because it is a population parameter.
From the sample, we can estimate σ^2 using the sample variance:

$$s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}.$$

The denominator $n-1$ makes s^2 an unbiased estimator of the population variance under the usual i.i.d. assumptions with finite variance.
When the population is normally distributed, replacing the unknown σ by the sample standard deviation s changes the reference distribution of the standardized mean from the standard normal distribution to Student's t distribution:

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}.$$

Here $n-1$ is the number of degrees of freedom.
Consequently, when σ is unknown and the population is normal, the exact two-sided confidence interval is

$$\bar{X} \pm t_{1-\alpha/2,\, n-1} \frac{s}{\sqrt{n}}.$$

For large samples, the Student's t distribution converges to the standard normal distribution, so the distinction between the two becomes small.
The degrees of freedom $n-1$ arise because the sample variance is estimated after the sample mean has been estimated from the same observations: one degree of freedom is lost.